

EXCERPT FROM TRIBULATION OF THE PRECURSOR

*Doing webpages and making apps is not Computer Science,
but a consequence and discipline in the very broad field of Computation*

Summary extract of the collection of private note fragments,
gathered and completed from GreyFox's Notebook

Shared for Jbaro Archives, marked by IIX.

January to March 2026

2026 // Open Classification

ABSTRACT

Four components of a single inquiry, developed across the period January to March 2026. *Biocomputers and Uploaded Intelligence* documents the engineering state of the field: which GPUs have been mapped, which platforms are running biological neural tissue, which algorithms exist and where they fall short. *The Algorithm of Who We Are* formalizes the identity system that must remain stable in any emulation attempt for the result to remain the same person, expressed as a system of differential equations, a fixed-point operator, and an optimal transport condition. *Computer Science* establishes the epistemic identity and core discipline CS is and why it is the one that has to solve this. *How Computer Science Borrowed from Neuroscience* traces the chain of debt from McCulloch in 1943 to the present day.

This excerpt summarizes each component and documents where they bear directly on each other, including where findings reinforce or contradict one another. No new claims are made. All references are taken from the source notebooks without modification.

1 Biocomputers and Uploaded Intelligence

The central constraint of the field is stated at the outset: mapping a connectome and executing it are two different problems. The connectome alone is not a nervous system: emulating every neuron, every synapse, every connection. Producing one requires electron microscopy at synaptic resolution, computational reconstruction, and annotation of cell types. As of 2024, complete connectomes exist for five organisms. The first was *C. elegans*, a nematode with 302 neurons, mapped in 1986 by White, Southgate, Thomson, and

Brenner [Bio-1]. By 2023, complete connectomes existed for four organisms: *C. elegans*, the tadpole larva of *Ciona intestinalis*, the marine annelid *Platynereis dumerilii*, and the *Drosophila* larva; the last comprising 3,016 neurons and 548,000 synapses, published by Winding et al. in *Science* [Bio-2]. In 2024, the connectome of an entire adult *Drosophila* brain was published by Dorkenwald et al.: approximately 140,000 neurons and tens of millions of synapses, annotated by cell type and predicted neurotransmitter identity [Bio-3]. This is the largest complete adult brain connectome produced to date.

On the mammalian side, a dataset released in 2025 by the MICrONS Consortium spans approximately one cubic millimeter of mouse visual cortex, integrating functional imaging of tens of thousands of neurons with electron microscopy reconstruction containing hundreds of thousands of cells and approximately half a billion synapses [Bio-4]. It is not a complete brain. It is a grain of sand. But it is a grain of sand at synaptic resolution. The scale gap is not a rounding error. It is the central engineering problem of the field.

Several platforms are currently attempting the execution side of this problem. FinalSpark's Neuroplatform, operational as of 2024, connects living human cortical organoids to multi-electrode arrays that allow both electrical stimulation and recording [Bio-5]. Organoids are clusters of lab-grown neurons derived from human stem cells that self-organize into layered structures resembling cortical tissue. Cortical Labs demonstrated in 2022 that cortical neurons cultured on a multi-electrode array could learn to play a simplified version of Pong using electrophysiological feedback, without being pre-programmed with the game's rules [Bio-6]. The neurons adapted their firing patterns in response to stimulation encoding the game state. The system learned from experience in a measurable behavioural sense. These platforms do not emulate a whole brain. They demonstrate that biological neural tissue can be kept alive outside the skull, interfaced with digital hardware, and made to produce computationally meaningful outputs. That is the engineering foundation on which any future emulation effort would have to build.

The Sandberg–Bostrom roadmap from 2008, produced through a workshop of invited experts at the Future of Humanity Institute, Oxford University, identified three core technical requirements for whole brain emulation: scanning the brain at sufficient resolution to capture the information that determines its function; translation (converting a structural scan into a computational model, determining which features are functionally relevant); and simulation (running the model on hardware at a speed and scale sufficient to produce real-time behaviour) [Bio-7]. The LOBE research literature also identifies embodiment as a formal requirement: providing the emulated system with sensory input and motor output so it can interact with an environment.

The 2025 State of Brain Emulation Report draws the critical formal distinction between simulation and emulation: simulation matches outputs, emulation reproduces the causal

machinery [Bio-8]. No current system has demonstrated emulation by that definition for any organism above *C. elegans*. The Blue Brain Project ran from 2005 to December 2024 under Henry Markram at EPFL, with the objective of building a biologically detailed computational model of the neocortical column of a rat [Bio-7]. Nineteen years on supercomputing resources. The result was electrophysiological simulation. Not emulation by Zanichelli's definition.

The distinction between simulation and emulation is not pedantic. It is the precise boundary between the two questions these materials are asking. The Sandberg–Bostrom roadmap, written in 2008, anticipated this distinction and stated it clearly. Seventeen years later, no system has crossed from simulation to emulation for any organism above 302 neurons.

The candidate algorithms for neural simulation exist in published form. The Hodgkin–Huxley model, published in 1952 and awarded the Nobel Prize in Physiology or Medicine in 1963, is the foundational algorithm for simulating how individual neurons generate and propagate electrical signals [Bio-9]. It expresses the firing mechanism of the action potential as a system of nonlinear differential equations governing sodium and potassium conductances across the cell membrane. The Integrate-and-Fire model (Lapicque 1907) is a useful simplification that captures the threshold-and-fire behaviour without modelling the full ionic dynamics [Bio-10]. The Sandberg–Bostrom roadmap describes a hierarchy of simulation levels, from simple rate models up through full electrophysiological models with protein concentrations and gene expression: higher integrity requires exponentially more data and compute [Bio-7]. A single modern GPU can simulate approximately 0.5 to 1 million neurons, with memory rather than compute as the primary bottleneck [Bio-8]. The primary limiting factor is identified as data, not hardware or algorithms.

“Brain emulation needs to take chemistry more into account than commonly occurs in current computational models. Chemical processes inside neurons have computational power on their own and occur on a vast range of timescales, from sub-millisecond to weeks.”

— Sandberg and Bostrom, *Whole Brain Emulation: A Roadmap*, 2008 [Bio-7]

2 The Algorithm of Who We Are

This component begins where Biocomputers stops. Connectome data and simulation algorithms address the structural problem. They do not address the integrity problem: what would need to be preserved for the result to still be the same person? The component formalises identity as a five-variable system:

$$\text{Identity}(p, t) := \{ A(p, t), R(p), C(p), S(p, t), M(p, t) \} \quad (1)$$

where A is attachment state, R is recursive self-reference, C is consciousness, S is the social network, and M is the autobiographical self. Each has a formal representation grounded in published research. Each raises a specific open question for the upload scenario.

Attachment (A)

A is the attachment state. Bowlby's 1969 monograph *Attachment* establishes that the attachment system is not a learned response to secondary drive but a primary motivational system with its own neural architecture [Alg-1]. The neurochemical substrate was documented by Insel and Young in 2001: oxytocin and vasopressin are the primary mediators of social attachment, operating through reward circuitry in the nucleus accumbens and ventral tegmental area [Alg-2]. Fisher, Aron, and Brown confirmed in 2005 that early-stage romantic love produces a measurable, reproducible neural signature in the caudate nucleus and VTA, consistent across subjects and distinct from sexual arousal [Alg-3]. The governing equation for A :

$$\frac{dA}{dt} = \lambda_{\text{form}} \cdot B(p, t) \cdot [\text{OXT}](p, t) - \lambda_{\text{decay}} \cdot A(t) - \delta(p, t) \cdot A(t) \quad (2)$$

where $B(p, t)$ is the dyadic bond function, $[\text{OXT}]$ is oxytocin concentration, and δ is an external disruption term. When $B(p, t) \rightarrow 0$ (the death of an attached person), the equation reduces to:

$$\frac{dA}{dt} = -(\lambda_{\text{decay}} + \delta) A(t) \implies A(t) = A_0 e^{-(\lambda + \delta)t} \quad (3)$$

Grief as exponential decay with time constant $\tau = 1/(\lambda + \delta)$. The second-order extension captures the dynamics of building and fading bonds over time:

$$\frac{d^2 A}{dt^2} = \lambda_{\text{form}} \left(\beta_1 \frac{dB}{dt} + \beta_2 \frac{d[\text{OXT}]}{dt} \right) - \lambda_{\text{decay}} \frac{dA}{dt} \quad (4)$$

Recursive self-reference (R)

R is recursive self-reference, formalised as a fixed-point problem on the Hilbert space of cognitive states. Hofstadter's argument in *Gödel, Escher, Bach* (1979) and *I Am a Strange Loop* (2007) is that the self is not a thing inside the brain but a process the brain runs: the process of modelling itself as an object in its own cognitive field [Alg-4, Alg-5]. The formal statement:

$$R^*(p) = F[R^*(p)] \quad (5)$$

where $F : L^2(\Omega) \rightarrow L^2(\Omega)$ and $\|F(R_1) - F(R_2)\| \leq k \|R_1 - R_2\|$, $k < 1$. By the Banach fixed-point theorem, a unique fixed point R^* exists and the iteration $R_{n+1} = F(R_n)$ converges to it from any starting point. The candidate form for the cognitive update operator is:

$$F[R](x) = \int_0^t K(x, x') R(x') dx' + \eta(x, A(p, t), M(p, t)) \quad (6)$$

where $K(x, x')$ is an attention kernel over the cognitive state space, and η is external input from the attachment and autobiographical memory components. F is a contraction if the operator norm of K is less than one. Open questions from the notebook: Is R^* unique for each person? Does the Lipschitz constant k vary with trauma? If the substrate changes at iteration n , does the sequence still converge to the same R^* ?

Consciousness (C)

C is consciousness, formalised via Tononi's Integrated Information Theory [Alg-8]:

$$\Phi(X) := \min_{M_k \in P(X)} \text{EI}(X | M_k) - H(X_t | X_{t-1}) - H(X_{t-1} | X_{M_k}) \quad (7)$$

$\Phi = 0$ means the system is fully decomposable with no integrated information. $\Phi > 0$ means irreducible causal structure. Whether any non-biological substrate running a brain map instantiates the same causal relationships is an open current research question. Baars' Global Workspace theory (1988) provides a complementary architectural constraint: consciousness requires information to be broadcast across a global workspace simultaneously available to a wide range of cognitive processes [Alg-9]. Chalmers' hard problem (1995) is the deepest obstruction: even after all functional problems are solved, no test currently exists that can be applied to a candidate upload and return a reliable answer about whether it is conscious [Alg-7].

Social network (S)

S is the social network, modelled as a weighted graph with Laplacian dynamics:

$$i\hbar \frac{\partial w_{ij}}{\partial t} = \gamma \cdot \text{interaction}(i, j, t) - \mu \cdot w_{ij}(t) \quad (8)$$

The graph Laplacian $L = D - W$ yields the Fiedler value $\lambda_2(L)$, algebraic connectivity, measuring social cohesion. Piolatto et al. 2022 confirmed that social isolation worsens

cognitive outcomes through combined mechanisms associated with measurable cognitive decline [Alg-10]. Post-upload, old edges freeze or decay: $dw_{ij}/dt = -\mu w_{ij}(t)$. The network collapses.

Autobiographical self (M)

M is the autobiographical self, formalised as a convolution integral over episodic memory, body state, and experience:

$$M(p, t) = \int_0^t K(t, \tau) [\text{mem}(p, \tau) + \text{body}(p, \tau) + \text{exp}(p, \tau)] d\tau \quad (9)$$

with exponential decay kernel $K(t, \tau) = e^{-\lambda(t-\tau)}$, following the Markov property. The notebook raises the alternative of a power-law kernel $K \sim (t - \tau)^{-\alpha}$ for long-range memory. Which kernel is cognitively faithful is an open question with direct computational implications.

Identity as trajectory

The identity-as-trajectory formulation is stated explicitly in the source:

$$p(t + dt) = p(t) + \frac{dp}{dt} dt + \mathcal{O}(dt^2) \quad (10)$$

where $dp/dt = f(A, R, C, S, M, \text{env}(t))$. Identity is a trajectory through state space, not a point. Uploading copies a snapshot at t_{upload} , not a trajectory. The formal integrity condition:

$$\text{Integrity}(p_{\text{upload}}) := \text{True iff } W_2(\mu_{\text{upload}}, \mu_{\text{orig}}) \leq \varepsilon \quad (11)$$

where W_2 is the Wasserstein-2 distance, the minimum cost of transporting one probability distribution over the identity state space into another. The upload problem is an optimal transport problem. $W_2 = 0$ requires perfect transport. By the continuity of W_2 and the irreversibility of substrate change, $L > 0$ almost surely. The total loss:

$$L_{\text{total}} = w_A L_A + w_R L_R + w_C L_C + w_S L_S + w_M L_M \quad (12)$$

where $L_x = \|x(p_{\text{upload}}, t) - x(p_{\text{orig}}, t)\|_2$ and the weights w_x are unknown.

“The equations in these pages are not solutions. They are the shape of the problem, written down so someone can see it clearly. The connectome gives us the wiring. It does not give us the loop, the bond, or the light inside.”

— Algorithm of Who We Are, conclusion

3 Computer Science

This component establishes the epistemological ground for the entire inquiry: what kind of discipline CS is, and why the upload problem belongs to it. The argument is that CS occupies a historically novel position among formal disciplines—simultaneously mathematics, engineering, and empirical science—as the direct consequence of one structural property that no other formal discipline possesses at scale.

CS began as formal mathematics. In 1936, Alan Turing published *On Computable Numbers* and defined a machine that existed only on paper, asking what it could in principle compute [CS-1]. Alongside Turing, Alonzo Church developed lambda calculus [CS-2] and John von Neumann formalised stored-program architecture [CS-3]. These were mathematicians drawing boundaries around a new formal object: computation itself. The name Computer Science was an institutional accident, applied when universities needed to house a department that fit neither Mathematics nor Engineering exactly.

The property that changed everything: the Turing machine, a purely theoretical construct, has a direct physical realisation in the transistor, the chip, the processor. When CS produces a formal system, that system can be executed. The abstraction does not merely describe a process. It becomes the process. No other formal discipline has this execution property at scale. The chain runs: logic to circuit to code to model to body to action.

When CS borrowed the McCulloch–Pitts model and built the Perceptron from it [CS-4, CS-5], it inherited the empirical assumptions of that model. Neural networks are not proven correct through formal proof. They are validated through testing against data from the real world. When the network gets a body mounted in a robot with cameras, force sensors, and actuators, the system must be tested empirically, against the resistance of the real world [CS-6]. The world pushes back. That is the defining feature of empirical science.

CS therefore occupies all three categories of rigorous knowledge simultaneously. It is mathematics when it produces formal proofs about algorithms and computability. It is engineering when it builds systems that must function under real-world physical constraints. It is empirical science when it trains models on data and validates them against observable reality. That triple nature is the source of its extraordinary reach: and it is what makes the upload problem a CS problem. Only CS can attempt to run a brain map. Only CS has the execution property that would make the attempt more than theoretical.

“Where would memories go? It becomes a philosophical problem.”

— GreyFox's Electrical Engineering Professor

4 How Computer Science Borrowed from Neuroscience

This component traces the historical chain by which CS arrived at the upload frontier. The chain begins not with an engineer but with two neurophysiologists. In 1943, Warren McCulloch and Walter Pitts published a paper attempting to describe, in formal logical terms, how a biological neuron produces an output from its inputs [Bor-1]. Their answer: the neuron is, in functional terms, a threshold gate. This was a description of neuroscience, not a proposal for engineering. The paper was published in the *Bulletin of Mathematical Biophysics*. It was read by engineers.

Frank Rosenblatt at the Cornell Aeronautical Laboratory was one of them. In 1958 he took the McCulloch–Pitts model and built physical hardware from it [Bor-2]. The Perceptron received multiple weighted input signals, summed them, and fired an output when the sum exceeded a threshold. The design was not derived from mathematical principles of computation. It was lifted directly from the neurophysiological description of nerve cells. The vocabulary of training, borrowed from animal learning research, entered CS. The vocabulary of neuroscience followed the hardware into the field and stayed there.

Minsky and Papert (1969) showed a single-layer perceptron could not solve problems that were not linearly separable, including the simple XOR operation [Bor-3]. Funding for connectionist research largely collapsed through the 1970s. In the 1980s, Rumelhart, Hinton, and Williams formalised backpropagation: error signals propagated backward through multiple layers, adjusting weights at each level proportionally [Bor-4]. The borrowed unit could now be stacked. By the 2010s, deep networks trained on large datasets and run on GPU hardware were outperforming every prior approach in image recognition, speech recognition, and machine translation [Bor-5]. Each layer remained, structurally, a population of threshold units derived from the McCulloch–Pitts neuron.

The Human Connectome Project, NIH-funded since 2009, has been working to chart the structural and functional connectivity of the human brain using diffusion MRI and functional neuroimaging [Bor-6]. FinalSpark's Neuroplatform connects living human cortical organoids to electrodes for both recording and stimulation [Bor-8]. DishBrain neurons learned Pong from feedback without pre-programmed rules [Bor-9]. These programs close the loop: the direction is now bidirectional.

Large language models trained on human-generated text were trained on essentially the entire recorded output of human language and thought. The architecture is biological in origin. The training signal is human in origin. The result is a system whose outputs were shaped at every stage by something that came from outside computer science.

“Maybe it’s almost as if we are trying to mirror ourselves and something is reflecting back?”

— GreyFox

5 Where the Components Connect

These are not four independent arguments. They are one argument approached from four angles, and reading them together surfaces connections that none of them resolve alone. Several of these connections constitute direct conflicts or complications that change the meaning of each component when read in isolation.

The simulation–emulation distinction from Biocomputers [Bio-8] cuts directly against the A integrity condition from the Algorithm component. The integrity condition is stated as a Wasserstein-2 distance over identity distributions: $W_2(\mu_{\text{upload}}, \mu_{\text{orig}}) \leq \varepsilon$. This metric measures the cost of transporting one probability distribution over behavioural and cognitive states into another. It is the right metric for identity-as-distribution. It is not the right metric for identity-as-causal-process. A system can match output distributions while failing the emulation condition.

The Algorithm component's loss function treats L_A and L_S as independent terms with unknown weights. The neurobiology cited within the same component does not support this independence. Insel and Young [Alg-2] document that attachment state A is mediated by oxytocin and vasopressin operating through the dopaminergic reward system. Piolatto et al. [Alg-10] document that social connectivity S protects cognition through neuroprotective pathways overlapping with the neurobiological effects of exercise. Both A and S are mediated by the same neurochemical substrate. Post-upload, the social network S decays as old edge weights follow $dw_{ij}/dt = -\mu w_{ij}$. As S decays, the biological drivers of A lose their input. L_A and L_S are coupled, not independent. The loss function as stated underestimates the damage that social graph collapse produces.

The CS component establishes that the only verification method available for any built system is empirical testing [Bor-6]. There is a structural incompatibility between what needs to be verified and the verification framework CS makes available: Chalmers' hard problem [Alg-7] is not a philosophical curiosity in this context. It is a direct obstruction to the engineering verification of the upload.

There is one further connection that reaches across all four components. The Borrowing component ends with the observation that large language models were trained on the recorded outputs of human minds. The CS component establishes that CS converts formal objects into systems instantiated in hardware. The Algorithm component asks what must be preserved for the upload to remain the same person. The Biocomputers component documents that we are already running human cortical tissue in computational loops outside the skull. Taken together: CS has been progressively importing more of the brain

into its architecture, then the outputs of human minds, then its neurobiological tissue. The upload question is not a future scenario. It is the logical conclusion of a trajectory that has been underway since 1943.

6 The Borrowing Trajectory and What It Implies for Uploaded Intelligence

The borrowing pattern documented in *How Computer Science Borrowed from Neuroscience* is toward the upload frontier without explicitly pursuing it. Each step borrowed more of the brain. The trajectory clarifies where the field actually stands with respect to uploaded intelligence.

The McCulloch–Pitts neuron (1943) was the first loan: the firing behaviour of a single neuron, abstracted as a threshold gate [Bor-1]. Nothing about memory, identity, or consciousness was claimed or attempted. Rosenblatt's Perceptron (1958) borrowed the learning mechanism through which neural tissue adapts [Bor-2]. Backpropagation and the multilayer network (1986) borrowed the cortex's hierarchical architecture [Bor-4]. Deep learning scaled this to dimensions that have no biological analogue, running on datasets and compute resources orders of magnitude larger than anything biological [Bor-5]. Large language models trained on human-generated text borrowed something harder to characterise: the outputs of human cognition at scale. Whether the process of generating that output was captured is precisely the question the Algorithm component asks.

Connectome mapping and biological computing platforms are attempting the next step: borrowing the wiring. The Human Connectome Project [Bor-6] maps structural and functional connectivity. The MICrONS dataset [Bio-4] reconstructs synaptic-resolution cortical structure. FinalSpark's Neuroplatform [Bio-5] and Cortical Labs' DishBrain [Bio-6, Bor-9] run biological neurons in computational loops. What is being borrowed at this stage: the actual neural architecture, connectivity, and tissue of biological brains—not a model of their outputs, but the thing itself.

The progression is: behaviour of one cell, then its learning mechanism, then its hierarchical architecture, then the outputs of human minds, then the wiring and tissue. The upload would be the limit of this sequence: not a model of the brain, not a copy of its architecture, not a simulation of its behaviour, but the brain's causal machinery reproduced with sufficient fidelity that the Wasserstein-2 distance between the original and the instantiation is within a tolerable ε . That limit has not been reached. The sequence is ongoing. The direction has not changed.

7 What This Amounts To

The question of uploaded intelligence is not a philosophical thought experiment. It is a formal engineering problem with documented components, published roadmaps, working platforms, and a mathematically precise integrity condition.

The hardware floor is known and expanding. Complete connectomes exist for five organisms up to 140,000 neurons [Bio-3]. Mammalian cortical reconstruction exists at synaptic resolution for a cubic millimetre of mouse visual cortex [Bio-4]. Living human neural tissue is being interfaced with digital hardware in real time [Bio-5, Bio-6]. The simulation algorithms are published and in active use [Bio-9, Bio-10]. A single modern GPU can simulate 0.5 to 1 million neurons. The primary limiting factor is physiological data, not hardware or algorithms [Bio-8].

The integrity problem is formally stated. Identity is a five-dimensional trajectory through state space: attachment A , recursive self-reference R , consciousness C , social network S , and autobiographical self M . Three specific blocking problems remain unresolved. The chemistry problem: no current simulation framework models neurochemical dynamics—specifically the [OXT] variable in the Algorithm's attachment equation—a data and modelling gap, not a hardware limitation [Bio-7]. The consciousness verification problem: Φ cannot be measured behaviourally, and the only verification method CS makes available is empirical behavioural testing [CS-6]. The trajectory problem: identity is a flow, not a snapshot. An upload captures a point on a trajectory. Whether the trajectory can restart on a new substrate and converge to the same attractor depends on whether F is a transferable structural property or an emergent property of feedback dynamics—a question the DishBrain result [Bio-6, Bor-9] bears on but does not resolve.

No current system satisfies the integrity condition. Not for any organism, and certainly not for a human. The Wasserstein distance between any uploaded mind and its original is greater than zero, and the tolerable ε is not known, and neither is who would decide it, or whether the person on the other side of the upload would be the one who agreed to it. These are not philosophical complications added after the fact. They follow directly from the formal structure of the integrity condition as stated in the source material. These are what the mathematics says when you read it carefully.

“The equations in these pages are not solutions. They are the shape of the problem, written down so someone can see it clearly. The connectome gives us the loop, the bond, or the light inside.”

Source notebooks: Biocomputers and Uploaded Intelligence; The Algorithm of Who We Are; Computer Science; How Computer Science Borrowed from Neuroscience. All from GreyFox's private notebook, shared for Jbaro Archives, January to March 2026.

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